

Important Questions for III internal

Subject : Mathematics.

Module III

Differential equation

1) Find the orthogonal trajectories of $\frac{x^2}{a^2} + \frac{y^2}{b^2 + \lambda} = 1$
Where λ is a parameter

Sol:-

Given:-

$$\frac{x^2}{a^2} + \frac{y^2}{b^2 + \lambda} = 1 \quad \text{--- (1)}$$

D.W.R.T 'x'

$$\frac{1}{a^2} (2x) + \frac{1}{b^2 + \lambda} (2y \cdot y_1) = 0$$

$\div 2$ on B.S.

$$\frac{x}{a^2} + \frac{y \cdot y_1}{b^2 + \lambda} = 0$$

$$\frac{x}{a^2} = -\frac{y \cdot y_1}{b^2 + \lambda} \quad \text{--- (2)}$$

from eqⁿ (1)

$$\frac{x^2}{a^2} = 1 - \frac{y^2}{b^2 + \lambda}$$

$$\Rightarrow \frac{x^2}{a^2} - 1 = -\frac{y^2}{b^2 + \lambda}$$

$$\Rightarrow \frac{x^2 - a^2}{a^2} = -\frac{y^2}{b^2 + \lambda} \quad \text{--- (3)}$$

Divide eqⁿ (2) by (3)

$$\frac{\frac{x}{a^2}}{\frac{x^2 - a^2}{a^2}} = \frac{\frac{y \cdot y_1}{b^2 + \lambda}}{\frac{-y^2}{b^2 + \lambda}}$$

$$\frac{x}{x^2 - a^2} = \frac{y_1}{y^2}$$

$\frac{x}{x^2 - a^2} = \frac{y_1}{y}$ — (4) is the diff. eqⁿ of the given family

Replace y_1 by $-\frac{1}{y_1}$ in eqⁿ (4)

$$\frac{x}{x^2 - a^2} = \frac{1}{y} \left(-\frac{1}{y_1} \right)$$

$$\frac{x}{x^2 - a^2} = \frac{1}{y} \left(-\frac{1}{\frac{dy}{dx}} \right)$$

$$\therefore \frac{x}{x^2 - a^2} = \frac{1}{y} \left(-\frac{dx}{dy} \right)$$

By variable separable

$$y \cdot dy = - \left(\frac{x^2 - a^2}{x} \right) \cdot dx$$

$$y \cdot dy = - \left(x + \frac{a^2}{x} \right) dx$$

Integrating

$$\frac{y^2}{2} = -\frac{x^2}{2} + a^2 \log x + c$$

$$\frac{x^2}{2} + \frac{y^2}{2} - a^2 \log x = c$$

multiply 2 on B.S.

$$x^2 + y^2 - 2a^2 \log x = 2c$$

$\therefore x^2 + y^2 - 2a^2 \log x = K$ // where $K = 2c$ is the req^d solⁿ

$$\frac{x^x}{x} = x$$

$$\frac{a^2}{x}$$

$$\int y \cdot dy = \frac{y^2}{2}$$

$$\int \frac{1}{x} dx = \log x$$

2. Show that the family of parabolas $y^2 = 4a(x+a)$ is self orthogonal

Given:-

$$y^2 = 4a(x+a) \quad \text{--- (1)}$$

D.w.r. to 'x'

$$2y \cdot y_1 = 4a(1+0)$$

$$2y \cdot y_1 = 4a$$

$\div 2$ on B.S.

$$y \cdot y_1 = 2a$$

$$\boxed{a = \frac{y y_1}{2}}$$

using $a = \frac{y y_1}{2}$ in eqⁿ (1)

$$y^2 = 4 \left(\frac{y y_1}{2} \right) \left(x + \frac{y y_1}{2} \right)$$

$$y^2 = 2(y y_1) \left(x + \frac{y y_1}{2} \right)$$

$$y^2 = 2x \cdot y y_1 + y^2 y_1^2$$

$\div y$ on B.S.

$$\boxed{y = 2x y_1 + y \cdot y_1^2} \quad \text{--- (2) is the differential eqⁿ of the given family of the curves}$$

Replace y_1 by $-\frac{1}{y_1}$

$$y = 2x \left(-\frac{1}{y_1} \right) + y \left(-\frac{1}{y_1} \right)^2$$

$$y = -\frac{2x}{y_1} + y \left(\frac{1}{y_1^2} \right)$$

multiply y_1^2 on B.S.

$$y y_1^2 = -2x y_1 + y$$

$$\boxed{y = 2x y_1 + y y_1^2} \quad \text{--- (3)}$$

from (2) & (3) the given family of the curves is said to be self orthogonal

3. Find the orthogonal trajectories of the family of curves $r^n = a^n \cos n\theta$.

⇒ take log on B.S.

$$\log r^n = \log a^n \cos n\theta \rightarrow \log r^n = \log a^n + \log (\cos n\theta)$$

$$n \log r = n \log a + \log (\cos n\theta)$$

D.W.r.t. θ

$$n \cdot \frac{1}{r} \frac{dr}{d\theta} = 0 + \frac{1}{\cos n\theta} (-\sin n\theta) \cdot n$$

÷ n on B.S.

$$\frac{1}{r} \frac{dr}{d\theta} = \frac{-1}{\cos n\theta} (-\sin n\theta)$$

$\frac{1}{r} \frac{dr}{d\theta} = -\tan n\theta$ is the Diff. eqn of the given curve

replace $\frac{dr}{d\theta}$ by $-r^2 \frac{d\theta}{dr}$

$$\frac{1}{r} \left(-r^2 \frac{d\theta}{dr} \right) = -\tan n\theta$$

$$r \frac{d\theta}{dr} = \tan n\theta$$

$$\frac{d\theta}{\tan n\theta} = \frac{dr}{r}$$

$$\frac{1}{\tan n\theta} = \cot n\theta$$

$$\boxed{\cot n\theta \cdot d\theta = \frac{dr}{r}} \quad \text{--- (1)}$$

∫ on B.S.

$$\int \cot n\theta \cdot d\theta = \int \frac{dr}{r} + \log b$$

$$\log \frac{\sin n\theta}{n} = \log r + \log b$$

$$\log \sin n\theta = n (\log r + \log b)$$

$$\log \sin n\theta = n \log (rb)$$

$$\cancel{\log} \sin n\theta = \cancel{\log} (rb)^n$$

$$\sin n\theta = r^n b^n$$

OR

from (i)

$$\cot n\theta \cdot d\theta = \frac{dr}{r}$$

$$\int \cot n\theta \cdot d\theta + \log b = \int \frac{dr}{r}$$

$$\log \frac{\sin n\theta}{n} + \log b = \log r$$

$$\frac{\log \sin n\theta + n \log b}{n} = \log r$$

$$\log (\sin n\theta) + \log b^n = n \log r$$

$$\log (\sin n\theta b^n) = \log r^n$$

$$\underline{\underline{b^n \sin n\theta = r^n}}$$

(4) Find the orthogonal trajectories of $r^n = a^n \sin n\theta$

taking log on B.S.

$$\log r^n = \log a^n \sin n\theta$$

$$n \log r = \log a^n + \log (\sin n\theta)$$

$$n \log r = n \log a + \frac{1}{\sin n\theta} (\cos n\theta) \cdot n$$

D.W. w.t. '0'

$$n \frac{1}{r} \frac{dr}{d\theta} = 0 + \frac{\cos n\theta}{\sin n\theta} \cdot n$$

Divide 'n' on B.S.

$$\frac{1}{r} \frac{dr}{d\theta} = \cot n\theta \quad \text{is the D.E of the given curve}$$

Replace $\frac{dr}{d\theta}$ by $-r^2 \frac{d\theta}{dr}$

$$\frac{1}{r} \left(-r^2 \frac{d\theta}{dr} \right) = \cot n\theta$$

$$-r \frac{d\theta}{dr} = \cot n\theta$$

$$\frac{d\theta}{\cot n\theta} = \frac{-dr}{r} = \tan n\theta \cdot d\theta = -\frac{dr}{r}$$

Int on B.S.
 $\int \tan n\theta \cdot d\theta =$

$$\frac{\log (\sec n\theta)}{n}$$

$$\frac{\log (\sec n\theta)}{n}$$

$$\frac{\log (\sec n\theta)}{n}$$

$$\log (\sec n\theta)$$

$$\sec n\theta$$

$$r^n =$$

$$r^n =$$

(5) A series
 and electric
 differential
 constant
 current
 $\Rightarrow$ the

$$\therefore IF = e^{\int P dx}$$

$\therefore$ The
 i.e.

OR

from ①

$$\cot n\theta \cdot d\theta = \frac{dr}{r}$$

$$\int \cot n\theta \cdot d\theta + \log b = \int \frac{dr}{r}$$

$$\log \frac{\sin n\theta}{n} + \log b = \log r$$

$$\frac{\log \sin n\theta + n \log b}{n} = \log r$$

$$\log (\sin n\theta) + \log b^n = n \log r$$

$$\log (\sin n\theta b^n) = \log r^n$$

$$\underline{\underline{b^n \sin n\theta = r^n}}$$

④ Find the orthogonal trajectories of $r^n = a^n \sin n\theta$
taking log on B.S.

$$\log r^n = \log a^n \sin n\theta$$

$$n \log r = \log a^n + \log (\sin n\theta)$$

$$n \log r = n \log a + \frac{1}{\sin n\theta} (\cos n\theta) \cdot n$$

D.W. w.r.t. θ

$$n \frac{1}{r} \frac{dr}{d\theta} = 0 + \frac{\cos n\theta}{\sin n\theta} \cdot n$$

Divide 'n' on B.S.

$$\frac{1}{r} \frac{dr}{d\theta} = \cot n\theta \quad \text{is the D.E of the given curve}$$

Replace $\frac{dr}{d\theta}$ by $-r^2 \frac{d\theta}{dr}$

$$\frac{1}{r} \left(-r^2 \frac{d\theta}{dr} \right) = \cot n\theta$$

$$-r \frac{d\theta}{dr} = \cot n\theta$$

$$\frac{d\theta}{\cot n\theta} = \frac{-dr}{r} = \tan n\theta \cdot d\theta \quad \text{--- } \frac{dr}{r}$$

$$\int \text{on B.S.}$$

$$\int \tan n\theta \cdot d\theta = - \int \frac{dr}{r}$$

$$\frac{\log(\sec n\theta)}{n} = -\log r + \log b$$

$$\frac{\log(\sec n\theta)}{n} = \log\left(\frac{b}{r}\right)$$

$$\frac{\log(\sec n\theta)}{n} = n \log\left(\frac{b}{r}\right)$$

$$\log(\sec n\theta) = \log\left(\frac{b}{r}\right)^n$$

$$\sec n\theta = \frac{b^n}{r^n}$$

$$r^n = \frac{b^n}{\sec n\theta}$$

$$r^n = b^n \cos n\theta$$

⑤ A series circuit with resistance R , inductance L and electromotive force E is governed by the differential eqⁿ $L \frac{di}{dt} + Ri = E$. Where L and R are constants and initially the current is zero. Find the current at any time t .

⇒ the D.E is $L \frac{di}{dt} + Ri = E$

÷ L on B.S

$$\left[\frac{di}{dt} + \frac{R}{L} i = \frac{E}{L} \right] \quad \text{--- P}$$

$$P = \frac{R}{L} \quad Q = \frac{E}{L}$$

$$\therefore IF = e^{\int R/L \cdot dt} = e^{R/L t}$$

∴ The general solution is

$$i \cdot e^{R/L t} = \int \frac{E}{L} e^{R/L t} \cdot dt + C$$

$$i.e. \frac{R}{L} t = \frac{E}{\cancel{L}} \frac{e^{\cancel{R/L} t}}{\cancel{R}} + c$$

$$\int e^{ax} dx = \frac{e^{ax}}{a}$$

$$i.e. \frac{R}{L} t = \frac{E}{R} e^{R/L \cdot t} + c$$

$\div e^{R/L \cdot t}$ on B.S.

$$i = \frac{E}{R} + \frac{c}{e^{R/L \cdot t}}$$

$$i = \frac{E}{R} + c e^{-R/L \cdot t} \quad \text{--- (1)}$$

When $t=0$, $i=0$

$$(1) \Rightarrow 0 = \frac{E}{R} + c \Rightarrow c = -\frac{E}{R}$$

Substitute 't' in eqn (1)

$$\therefore i = \frac{E}{R} - \frac{E}{R} e^{-R/L \cdot t}$$

$$i = \frac{E}{R} (1 - e^{-R/L \cdot t})$$

(6) The L-R circuit diff. eqn acted on by an electromotive force $E \sin \omega t$ satisfies the DE $L \frac{di}{dt} + Ri = E \sin \omega t$. If there is no current in the circuit initially, obtain the value of current at any time t .

Soln:-

$$\text{Given } L \frac{di}{dt} + Ri = E \sin \omega t$$

$\div L$ on B.S.

$$\frac{di}{dt} + \frac{R}{L} i = \frac{E}{L} \sin \omega t \quad \text{is linear Diff. eqn}$$

$$P = \frac{R}{L} \quad Q = \frac{E}{L} \sin \omega t$$

$$IF = e^{\int P \cdot dt} = e^{\int \frac{R}{L} dt} = e^{R/L t}$$

General solution is.

$$i \times I.F = \int (Q \times IF) dt + c$$

$$i \cdot e^{R/L t} = \int \left(\frac{E}{L} \sin \omega t e^{R/L t} \right) dt + c$$

To solve above integrals use the following formula

$$\int e^{at} \sin bt = \frac{e^{at}}{\sqrt{a^2 + b^2}} \sin [bt - \tan^{-1}(b/a)]$$

$$\text{Here } a = \frac{R}{L} \quad b = \omega$$

$$i \cdot e^{R/L t} = \frac{E}{L}$$

$$i \cdot e^{R/L t} = \frac{E}{L} \left[\frac{e^{R/L t}}{\sqrt{\left(\frac{R}{L}\right)^2 + (\omega)^2}} \left(\sin \left(\omega t - \tan^{-1} \left(\frac{\omega}{R/L} \right) \right) \right) + c \right]$$

$$i \cdot e^{R/L t} = \frac{E}{L} \left[\frac{e^{R/L t}}{\sqrt{\frac{R^2}{L^2} + \omega^2}} \left(\sin \left(\omega t - \tan^{-1} \left(\frac{\omega L}{R} \right) \right) \right) + c \right]$$

$\div e^{R/L t}$ on B.S

$$i = \frac{E}{L} \left[\frac{1}{\sqrt{\frac{R^2 + \omega^2 L^2}{L^2}}} \left\{ \sin \left(\omega t - \tan^{-1} \left(\frac{\omega L}{R} \right) \right) \right\} + \frac{c}{e^{R/L t}} \right]$$

$$P \tan^{-1} \left(\frac{\omega L}{R} \right) = \phi$$

$$\therefore i = \frac{E}{L} \left(\frac{L}{\sqrt{R^2 + \omega^2 L^2}} \left\{ \sin (\omega t - \phi) \right\} + c e^{-R/L t} \right) \quad \text{--- (1)}$$

It is given that when $t=0$, $i=0$

$$0 = \frac{E}{\sqrt{R^2 + \omega^2 L^2}} \left\{ \sin (-\phi) \right\} + c$$

$$0 = \frac{-E}{\sqrt{R^2 + \omega^2 L^2}} \sin \phi + c \quad \sin(-0) = -\sin 0$$

$$C = \frac{E}{\sqrt{R^2 + \omega^2 L^2}} \sin \phi$$

Substitute C in eqⁿ (1)

$$i = \frac{E}{\sqrt{R^2 + \omega^2 L^2}} \{ \sin(\omega t - \phi) \} + \frac{E}{\sqrt{R^2 + \omega^2 L^2}} \sin \phi e^{-R/L t}$$

$$i = \frac{E}{\sqrt{R^2 + \omega^2 L^2}} \left[\sin(\omega t - \phi) + \sin \phi e^{-R/L t} \right]$$

where $\phi = \tan^{-1} \left(\frac{\omega L}{R} \right)$

7) Solve $xy \left(\frac{dy}{dx} \right)^2 - (x^2 + y^2) \frac{dy}{dx} + xy = 0$

Solⁿ: Let $p = \frac{dy}{dx}$

$$xy p^2 - (x^2 + y^2) p + xy = 0$$

$$ax^2 + bx + c = 0$$

$$a = xy \quad b = -(x^2 + y^2) \quad c = xy$$

$$p = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$p = \frac{(x^2 + y^2) \pm \sqrt{(-x^2 + y^2)^2 - 4(xy)(xy)}}{2(xy)}$$

$$p = \frac{(x^2 + y^2) \pm \sqrt{(x^2 + y^2)^2 - 4(xy)(xy)}}{2xy}$$

$$p = \frac{x^2 + y^2 \pm \sqrt{(x^2)^2 + (y^2)^2 + 2x^2 y^2 - 4x^2 y^2}}{2xy}$$

$$p = \frac{x^2 + y^2 \pm \sqrt{(x^2)^2 + (y^2)^2 - 2x^2 y^2}}{2xy}$$

$$P = \frac{x^2 + y^2 \pm \sqrt{(x^2 + y^2)^2}}{2xy}$$

$$P = \frac{x^2 + y^2 \pm (x^2 - y^2)}{2xy}$$

$$P = \frac{x^2 + y^2 + (x^2 - y^2)}{2xy}$$

(or)

$$P = \frac{x^2 + y^2 - (x^2 - y^2)}{2xy}$$

$$P = \frac{x^2 + \cancel{y^2} + x^2 - \cancel{y^2}}{2xy}$$

(or)

$$P = \frac{\cancel{x^2} + y^2 - \cancel{x^2} + y^2}{2xy}$$

$$P = \frac{x \cancel{x}^1}{x \cancel{y}^1}$$

(or)

$$P = \frac{\cancel{y}^1}{\cancel{x}^1}$$

$$P = \frac{x}{y}$$

(or)

$$P = \frac{y}{x}$$

$$\frac{dy}{dx} = \frac{x}{y}$$

(or)

$$\frac{dy}{dx} = \frac{y}{x}$$

$$y \cdot dy = x \cdot dx$$

(or)

$$\frac{dy}{y} = \frac{dx}{x}$$

Integrating

$$\frac{y^2}{2} = \frac{x^2}{2} + \frac{C}{2}$$

$$\log y = \log x + \log C$$

$$\log y = \log (x \cdot C)$$

$$\frac{y^2}{2} - \frac{x^2}{2} - \frac{C}{2} = 0$$

$$y = x \cdot C$$

$$y - xC = 0$$

$$y^2 - x^2 - C = 0$$

∴ The general solⁿ is $(y^2 - x^2 - C)(y - xC) = 0$

(8) solve $y \left(\frac{dy}{dx} \right)^2 + (x - y) \frac{dy}{dx} - x = 0$

Solⁿ

$$yp^2 + (x-y)p - x = 0$$

$$ax^2 + bx + c = 0$$

$$a=y \quad b=(x-y) \quad c=-x$$

$$P = \frac{-(x-y) \pm \sqrt{(x-y)^2 - 4(y)(-x)}}{2(y)}$$

$$P = \frac{-(x-y) \pm \sqrt{x^2 + y^2 - 2xy + 4xy}}{2y}$$

$$P = \frac{-(x-y) \pm \sqrt{x^2 + y^2 + 2xy}}{2y}$$

$$P = \frac{-(x-y) \pm \sqrt{(x+y)^2}}{2y}$$

$$P = \frac{-(x-y) \pm (x+y)}{2y}$$

$$P = \frac{-(x-y) + (x+y)}{2y}$$

$$\textcircled{os} \quad P = \frac{-(x-y) - (x+y)}{2y}$$

$$P = \frac{-\cancel{x} + y + \cancel{x} + y}{2y}$$

$$\textcircled{os} \quad P = \frac{-\cancel{x} + \cancel{y} - \cancel{x} - \cancel{y}}{2y}$$

$$P = \frac{\cancel{2y}}{\cancel{2y}} = 1$$

$$\textcircled{os} \quad P = \frac{-\cancel{2x}}{\cancel{2y}} = \frac{-x}{y}$$

$$\frac{dy}{dx} = 1$$

$\textcircled{os}$

$$\frac{dy}{dx} = \frac{-x}{y}$$

$$dy = dx$$

$\textcircled{os}$

$$y \cdot dy = -x \cdot dx$$

Integrating

$$y = x + C$$

$$\boxed{y - x - C = 0}$$

$$\frac{y^2}{2} = -\frac{x^2}{2} + \frac{C}{2}$$

$$\frac{y^2}{2} + \frac{x^2}{2} - \frac{C}{2} = 0$$

multiply 2 on B.S

$$x^2 + y^2 - C = 0$$

$\therefore$ General solⁿ is $(y-x-c)(x^2-y^2-c)=0$

q) Solve the eqⁿ $(px-y)(py+x)=2p$ by reducing into clairaut's eqⁿ taking the substitution $x=x^2$, $y=y^2$
Given:

$$\begin{aligned}x &= x^2 & , & & y &= y^2 \\dx &= 2x \cdot dx & , & & dy &= 2y \cdot dy \\ \frac{dx}{dx} &= 2x & , & & \frac{dy}{dy} &= 2y\end{aligned}$$

Now,

$$p = \frac{dy}{dx} \quad \frac{dy}{dx} \quad \frac{dx}{dx}$$

$$p = \frac{1}{2}y \quad p = 2x$$

$$\boxed{p = \frac{x}{y} p}$$

Substitute p in given eqⁿ

$$\left(\frac{x}{y} p x - y\right) \left(\frac{x}{y} p y + x\right) = 2 \left(\frac{x}{y} p\right)$$

$$\left(\frac{x^2 p}{y} - y\right) (xp + x) = 2 \frac{x}{y} p$$

$$\left(\frac{x^2 p - y^2}{y}\right) x (p+1) = 2 \frac{x}{y} p$$

$$(xp - y) (p+1) = 2p$$

$$(xp - y) = \frac{2p}{p+1}$$

$$\boxed{y = px - \frac{2p}{p+1}}$$

$\therefore$ The general solⁿ is.

$$y = cx - \frac{2c}{c+1}$$

$$\boxed{y^2 = cx^2 - \frac{2c}{c+1}}$$

10) solve the eqⁿ $(px-y)(py+x) = a^2p$ by taking
 $x = x^2$ $y = y^2$

$$\frac{dx}{dy} = 2x$$

$$\frac{dy}{dx} = 2y$$

Now,

$$p = \frac{dy}{dx}$$

$$\frac{dy}{dx}$$

$$\frac{dx}{dx}$$

$$p = \frac{dy}{dx}$$

$$p = \frac{1}{2y} \quad p \propto x$$

$$p = \frac{x}{y} p$$

Substitute p in given eqⁿ

$$(px-y)(px+x) = a^2p$$

$$\left(\frac{x}{y} px - y\right) \left(\frac{x}{y} py + x\right) = a^2 \left(\frac{x}{y} p\right)$$

$$\left(\frac{x^2}{y} p - y\right) (xp + x) = a^2 \frac{x}{y} p$$

$$\left(\frac{x^2 p - y}{y}\right) x (p+1) = a^2 \frac{x}{y} p$$

$$\left(\frac{xp - y}{y}\right) x (p+1) = a^2 \frac{x}{y} p$$

$$(xp - y)(p+1) = a^2 p$$

$$xp - y = \frac{a^2 p}{p+1}$$

$$y = px - \frac{a^2 p}{p+1} \quad \text{is in Clairaut's form}$$

$\therefore$ The general solⁿ is

$$y = cx = \frac{a^2 c}{c+1}$$

$$y^2 = cx^2 - \frac{a^2 c}{c+1}$$

Module-5

Linear Algebra

① Find the rank of the matrix $A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$

$$R_1 \leftrightarrow R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_1$$

$$R_4 \rightarrow R_4 - R_1$$

$$\sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$R_4 \rightarrow R_4 - R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\rho(A) = 2$$

2) Test for consistency and solve:

~~$$x + y + z = 6$$~~

~~$$x - 2y + 3z = 8$$~~

~~$$2x + y + z = 3$$~~

$$x + y + z = 6$$

$$x - y + 2z = 5$$

$$3x + y + z = 8$$

Soln.

$$[A:B] = \begin{bmatrix} 1 & 1 & 1:6 \\ 1 & -1 & 2:5 \\ -3 & 1 & 1:8 \end{bmatrix}$$

$$R_2' \rightarrow R_2 - R_1$$

$$R_3' \rightarrow R_3 - 3R_1$$

$$\sim \begin{bmatrix} 1 & 1 & 1:6 \\ 0 & -2 & 1:-1 \\ 0 & -2 & -2:-10 \end{bmatrix}$$

$$R_3' \rightarrow R_3 - R_2$$

$$\sim \begin{bmatrix} 1 & 1 & 1:6 \\ 0 & -2 & 1:-1 \\ 0 & 0 & -3:-9 \end{bmatrix}$$

$$R_3' \rightarrow -\frac{1}{3} R_3$$

$$\sim \begin{bmatrix} 1 & 1 & 1:6 \\ 0 & -2 & 1:-1 \\ 0 & 0 & 1:3 \end{bmatrix}$$

$$R_2' \rightarrow -\frac{1}{2} R_2$$

$$\sim \begin{bmatrix} 1 & 1 & 1:6 \\ 0 & 1 & -\frac{1}{2}:\frac{1}{2} \\ 0 & 0 & 1:3 \end{bmatrix}$$

$$\rho(A) = 3$$

$$\rho(A:B) = 3$$

$$\therefore \rho(A) = \rho(A:B) = 3 = n \text{ \& } n = 3$$

$\therefore$ The given system has unique solution.

∴ Given matrix is equivalent to $x + y + z = 6$

$$y - \frac{1}{2}z = \frac{1}{2}$$

$$z = 3$$

By Backward Substitution

$$\textcircled{2} \Rightarrow y - \frac{1}{2}(3) = \frac{1}{2}$$

$$y = \frac{1}{2} + \frac{3}{2}$$

$$y = \frac{4}{2} = 2$$

$$\boxed{y = 2}$$

$$\textcircled{1} \quad x + 2 + 3 = 6$$

$$x + 5 = 6$$

$$x = 6 - 5$$

$$\boxed{x = 1}$$

$$\therefore x = 1, y = 2, z = 3$$

(3) Solve the following system of eq^{ns} by Gauss elimination method.

$$x + y + z = 9$$

$$x - 2y + 3z = 8$$

$$2x + y - z = 3$$

Soln:- The Augmented matrix

$$[A : B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 1 & -2 & 3 & 8 \\ 2 & 1 & -1 & 3 \end{array} \right]$$

$$R_2' \rightarrow R_2 - R_1$$

$$R_3' \rightarrow R_3 - 2(R_1)$$

$$\sim \begin{bmatrix} 1 & 1 & 1:9 \\ 0 & -3 & 2:-1 \\ 0 & -1 & -3:-15 \end{bmatrix}$$

$$R_2' \rightarrow -\frac{1}{3}R_2$$

$$\sim \begin{bmatrix} 1 & 1 & 1:9 \\ 0 & 1 & -2/3:1/3 \\ 0 & -1 & -3:-15 \end{bmatrix}$$

$$R_3' \rightarrow R_3 + R_2$$

$$\sim \begin{bmatrix} 1 & 1 & 1:9 \\ 0 & 1 & -2/3:1/3 \\ 0 & 0 & -11/3:-44/3 \end{bmatrix}$$

$$R_3' \rightarrow -\frac{3}{11}R_3$$

$$\sim \begin{bmatrix} 1 & 1 & 1:9 \\ 0 & 1 & -2/3:1/3 \\ 0 & 0 & 1:4 \end{bmatrix}$$

This is equivalent to $x+y+z=9$

$$y - \frac{2}{3}z = \frac{1}{3}$$

$$\boxed{z=4}$$

By back substitution

$$y - \frac{2}{3}(4) = \frac{1}{3}$$

$$y - \frac{8}{3} = \frac{1}{3}$$

$$y = \frac{1}{3} + \frac{8}{3}$$

$$y = \frac{1+8}{3} = \frac{9}{3} = 3$$

$$x+3+4=9$$

$$x+7=9$$

$$x=9-7$$

$$\boxed{x=2}$$

$$\boxed{y=3}$$

$$\boxed{z=4}$$

4.) Apply Gauss Jordan method to solve the system of equations

$$2x+5y+7z=52$$

$$2x+y-z=0$$

$$x+y+z=9$$

$$[A:B] = \begin{bmatrix} 1 & 1 & 1 : 9 \\ 2 & 1 & -1 : 0 \\ 2 & 5 & 7 : 52 \end{bmatrix}$$

$$R_2' \rightarrow R_2 - 2R_1$$

$$R_3' \rightarrow R_3 - 2R_1$$

$$\hookrightarrow \begin{bmatrix} 1 & 1 & 1 : 9 \\ 0 & -1 & -3 : -18 \\ 0 & 3 & 5 : 34 \end{bmatrix}$$

$$R_2' \rightarrow -R_2$$

$$\hookrightarrow \begin{bmatrix} 1 & 1 & 1 : 9 \\ 0 & 1 & 3 : 18 \\ 0 & 3 & 5 : 34 \end{bmatrix}$$

$$R_3' \rightarrow R_3 - 3(R_2)$$

$$\hookrightarrow \begin{bmatrix} 1 & 1 & 1 : 9 \\ 0 & 1 & 3 : 18 \\ 0 & 0 & -4 : -20 \end{bmatrix}$$

$$R_3' \rightarrow -\frac{1}{4}R_3$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & 9 \\ 0 & 1 & 3 & : & 18 \\ 0 & 0 & 1 & : & +5 \end{bmatrix}$$

$$R_1' \rightarrow R_1 - R_2$$

$$\sim \begin{bmatrix} 1 & 0 & -2 & : & -9 \\ 0 & 1 & 3 & : & 18 \\ 0 & 0 & 1 & : & 5 \end{bmatrix}$$

$$R_1' \rightarrow R_1 + 2R_3$$

$$R_2' \rightarrow R_2 - 3R_3$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & : & 1 \\ 0 & 1 & 0 & : & 3 \\ 0 & 0 & 1 & : & 5 \end{bmatrix}$$

$$x=1, y=3, z=5$$

(5) solve the following system of eqⁿ by Gauss siedel method

$$10x + y + z = 12$$

$$x + 10y + z = 12$$

$$x + y + 10z = 12$$

The given system is diagonally dominant

$$10x + y + z = 12 \Rightarrow 10x = 12 - y - z$$

$$x = \frac{1}{10}(12 - y - z)$$

$$x + 10y + z = 12 \Rightarrow 10y = 12 - x - z$$

$$y = \frac{1}{10} [12 - x - z]$$

$$x + y + 10z = 12 \Rightarrow 10z = 12 - x - y$$

$$z = \frac{1}{10} [12 - x - y]$$

Take $x^{(0)} = 0$, $y^{(0)} = 0$, $z^{(0)} = 0$.

First iteration

$$z^{(1)} = \frac{1}{10} (12 - 0 - 0) = \frac{12}{10} = 1.2$$

$$y^{(1)} = \frac{1}{10} (12 - 1.2 - 0) = \frac{10.8}{10} = 1.08$$

$$x^{(1)} = \frac{1}{10} (12 - 1.2 - 1.08) = 0.972$$

Second iteration

$$z^{(2)} = \frac{1}{10} (12 - 1.08 - 0.972) = 0.9948$$

$$y^{(2)} = \frac{1}{10} (12 - 0.9948 - 0.972) = 1.0033$$

$$x^{(2)} = \frac{1}{10} (12 - 0.9948 - 1.0033) = 1.0001$$

Third iteration

$$z^{(3)} = \frac{1}{10} (12 - 1.0033 - 1.0001) = 0.9996 \approx 1$$

$$y^{(3)} = \frac{1}{10} (12 - 0.9996 - 1.0001) = 1.00003$$

$$x^{(3)} = \frac{1}{10} (12 - 0.9996 - 1.00003) = 1.00004$$

Fourth iteration

$$x = \frac{1}{10}(12 - 1 - 1) = \frac{10}{10} = 1$$

$$y = \frac{1}{10}(12 - 1 - 1) = \frac{10}{10} = 1$$

$$z = \frac{1}{10}(12 - 1 - 1) = \frac{10}{10} = 1$$

$$10x + y + z = 12$$

$$10(1) + 1 + 1 = 12$$

(6) Find the largest eigen value and the corresponding eigen vector of the matrix A by the power method given that

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

Solⁿ $X^{(0)} = [1 \ 0 \ 0]^T$

$$A X^{(0)} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \\ 0.5 \end{bmatrix} = \lambda_1 X^{(1)}$$

$$A X^{(1)} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 2.5 \\ 0 \\ 2 \end{bmatrix} = 2.5 \begin{bmatrix} 1 \\ 0 \\ 0.8 \end{bmatrix} = \lambda_2 X^{(2)}$$

$$A X^{(2)} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.8 \end{bmatrix} = \begin{bmatrix} 2.8 \\ 0 \\ 2.6 \end{bmatrix} = 2.8 \begin{bmatrix} 1 \\ 0 \\ 0.92 \end{bmatrix} = \lambda_3 X^{(3)}$$

$$AX^{(3)} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.92 \end{bmatrix} = \begin{bmatrix} 2.92 \\ 0 \\ 2.84 \end{bmatrix} = 2.92 \begin{bmatrix} 1 \\ 0 \\ 0.9726 \end{bmatrix} = \lambda_2 X^{(4)}$$

$$AX^{(4)} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.97 \end{bmatrix} = \begin{bmatrix} 2.97 \\ 0 \\ 2.94 \end{bmatrix} = 2.97 \begin{bmatrix} 1 \\ 0 \\ 0.989 \end{bmatrix} = \lambda_5 X^{(5)}$$

$$AX^{(5)} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.989 \end{bmatrix} = \begin{bmatrix} 2.989 \\ 0 \\ 2.978 \end{bmatrix} =$$

$$2.989 \begin{bmatrix} 1 \\ 0 \\ 0.996 \end{bmatrix} = \lambda_6 (X)^{(6)}$$

$\therefore$ The largest eigen value = 2.989 ≈ 3 and the corresponding eigen vector $x = [0.996]^T \approx [1 \ 0 \ 1]^T$

(7) Investigate the values of λ & μ such that the system of eq^{ns}

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$x + 2y + \lambda z = \mu \text{ may have}$$

(a) unique solⁿ

(b) infinite solⁿ

(c) No solⁿ

Solⁿ:-

$$[A : B] = \begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{bmatrix}$$

$$R_2' \rightarrow R_2 - R_1$$

$$R_3' \rightarrow R_3 - R_1$$

$$\sim \begin{bmatrix} 1 & 1 & 1:6 \\ 0 & 1 & 2:4 \\ 0 & 1 & \lambda-1: \mu-6 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\sim \begin{bmatrix} 1 & 1 & 1:6 \\ 0 & 1 & 2:4 \\ 0 & 0 & \lambda-3: \mu-10 \end{bmatrix}$$

(a) unique solution:

The system has unique solⁿ only when $\lambda-3 \neq 0$ and whatever the value of μ

(b) Infinite solⁿ:

The system has infinite solⁿ only when $\lambda-3=0$ and $\mu-10=0$

$$\therefore \boxed{\lambda=3} \text{ and } \boxed{\mu=10}$$

(c) No solⁿ:

The system has no solⁿ only when $\lambda-3=0$ and $\mu-10 \neq 0$

$$\therefore \boxed{\lambda=3} \text{ and } \boxed{\mu \neq 10}$$

(8) solve $(x^2+y^2+z) dx + xy dy = 0$.

→ solⁿ: $M = x^2 + y^2 + x$ $N = xy$

$$\frac{\partial M}{\partial y} = 2y$$

$$\frac{\partial N}{\partial x} = y$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x} = \text{Given D.E is not exact}$$

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 2y - y$$

$$= +y$$

which is nearest to N

$$\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{y}{xy} = \frac{1}{x} = f(x)$$

$$\int f(x) \cdot dx$$
$$I.F = e^{\int \frac{1}{x} \cdot dx}$$

$$I.F = e^{\log x}$$

$$I.F = x$$

$$I.F = x$$

multiply $I.F=x$ in the given D.E

$$x(x^2 + xy^2 + x)dx + (xy dy) = 0$$

$$x^3 + xy^2 + x^2 dx + x^2 y dy = 0$$

$$M = x^3 + xy^2 + x^2 \quad N = x^2 y$$

$$\frac{\partial M}{\partial y} = 2xy$$

$$\frac{\partial N}{\partial x} = 2xy$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow \text{D.E is Exact.}$$

$\therefore$ solⁿ is

$$\int M \cdot dx = \int N(y) \cdot dy = C$$

$$\int (x^3 + xy^2 + x^2) dx + \int 0 dy = 0$$

$$\frac{x^4}{4} + y^2 \frac{x^2}{2} + \frac{x^3}{3} = C$$